

9 Three planes have equations

$$kx + y - 2z = 0$$

$$2x + 3y - 6z = -5$$

$$3x - 2y + 5z = 1$$

where k is a constant.

Investigate the arrangement of the planes for each of the following cases. If in either case the planes meet at a unique point, find the coordinates of that point.

(a) $k = -1$ **[3]**

(b) $k = \frac{2}{3}$ **[4]**